

# 1D CNN — Architecture Variant Comparison

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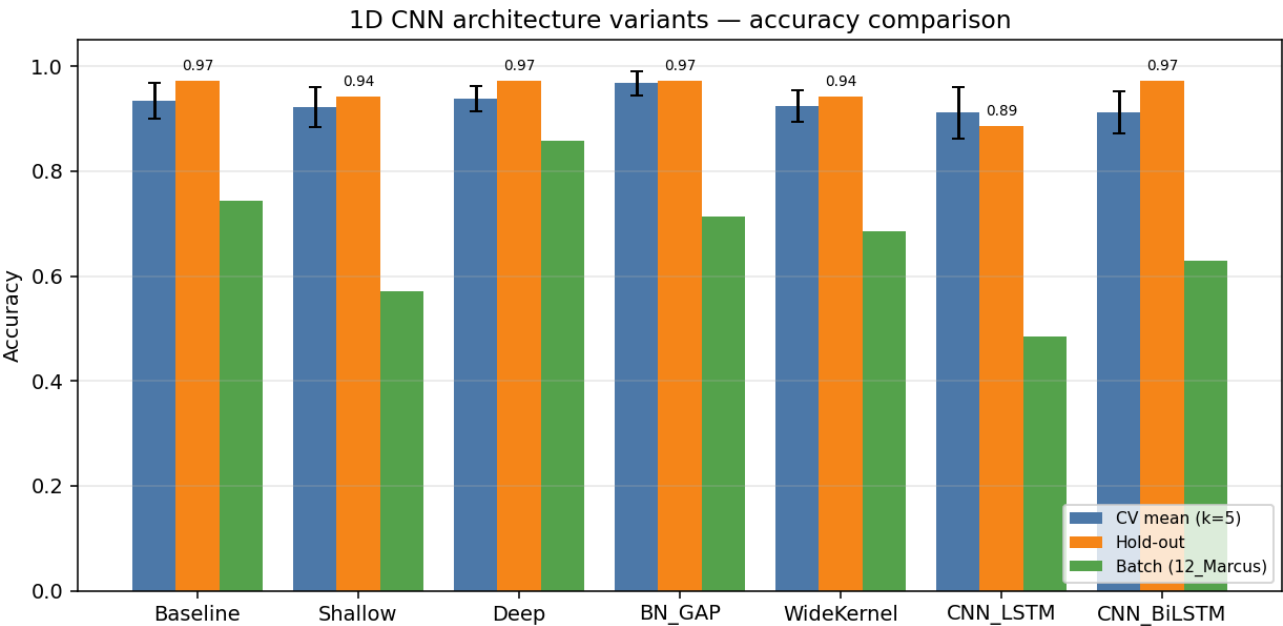
## Data & configuration

|                     |                                                                                                             |
|---------------------|-------------------------------------------------------------------------------------------------------------|
| Training path 1     | /home/jestin/ThesisRepo/ML/NewTestData/1_Alan/Dynamic                                                       |
| Training path 2     | /home/jestin/ThesisRepo/ML/NewTestData/2_Alex/Dynamic                                                       |
| Training path 3     | /home/jestin/ThesisRepo/ML/NewTestData/3_Anghad/Dynamic                                                     |
| Training path 4     | /home/jestin/ThesisRepo/ML/NewTestData/4_Daniel/Dynamic                                                     |
| Training path 5     | /home/jestin/ThesisRepo/ML/NewTestData/6_Harry/Dynamic                                                      |
| Training path 6     | /home/jestin/ThesisRepo/ML/NewTestData/8_Jestin/Dynamic                                                     |
| Training path 7     | /home/jestin/ThesisRepo/ML/NewTestData/11_Mansh/Dynamic                                                     |
| Training path 8     | /home/jestin/ThesisRepo/ML/NewTestData/14_StephenV2/Dynamic                                                 |
| Training path 9     | /home/jestin/ThesisRepo/ML/NewTestData/15_Tash/Dynamic                                                      |
| Training path 10    | /home/jestin/ThesisRepo/ML/NewTestData/17_Raquel/Dynamic                                                    |
| Batch-test root     | /home/jestin/ThesisRepo/ML/NewTestData/12_Marcus/Dynamic                                                    |
| Total trials        | 341 (train pool 306, hold-out 35)                                                                           |
| Classes             | 7: Double_Lower, Double_Nothing, Double_Pistol_Recoil, Double_Raise, Double_Wiggle, Double_cmere, Drum_Roll |
| Sensor channels     | 176 (sequence length 90)                                                                                    |
| Resample / filter   | 90 steps · Butterworth cutoff 15.0 Hz, order 4                                                              |
| Normalisation       | minmax                                                                                                      |
| CV folds · shuffle  | 5 · True                                                                                                    |
| Augmentation        | on · copies=5 · amplitude_scale, baseline_offset                                                            |
| Epochs · batch size | 15 · 16                                                                                                     |

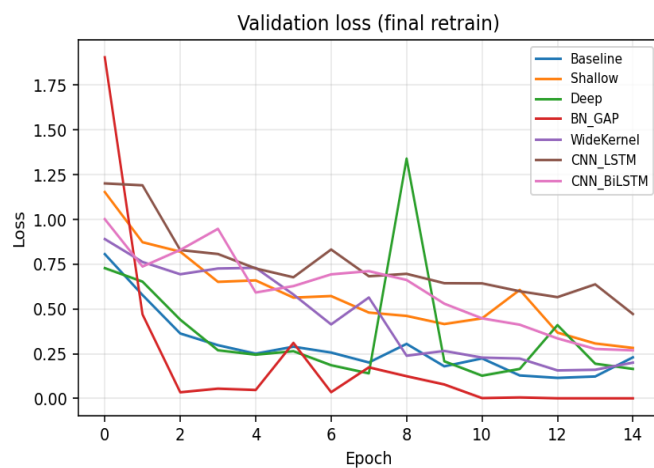
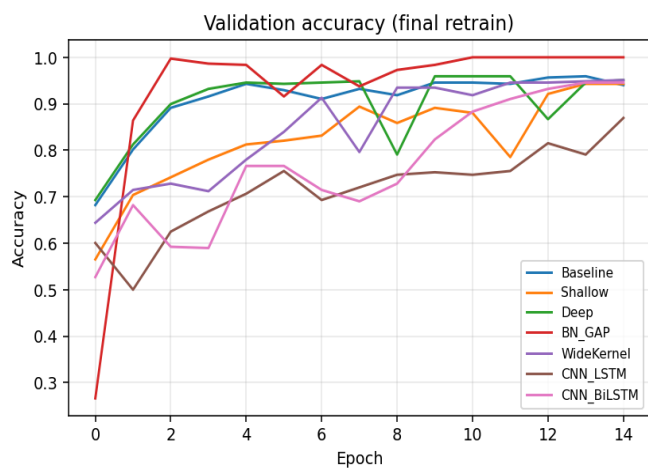
## Variant comparison (summary)

| Variant    | Params  | CV mean (k=5) | CV std | Hold-out acc | Hold-out loss | Batch acc | Batch n |
|------------|---------|---------------|--------|--------------|---------------|-----------|---------|
| Baseline   | 113,255 | 0.9346        | 0.0345 | 0.9714       | 0.0692        | 0.7429    | 26/35   |
| Shallow    | 72,487  | 0.9216        | 0.0379 | 0.9429       | 0.2130        | 0.5714    | 20/35   |
| Deep       | 204,007 | 0.9380        | 0.0239 | 0.9714       | 0.1302        | 0.8571    | 30/35   |
| BN_GAP     | 57,319  | 0.9673        | 0.0232 | 0.9714       | 0.1233        | 0.7143    | 25/35   |
| WideKernel | 131,687 | 0.9249        | 0.0304 | 0.9429       | 0.1480        | 0.6857    | 24/35   |
| CNN_LSTM   | 68,455  | 0.9117        | 0.0494 | 0.8857       | 0.3586        | 0.4857    | 17/35   |
| CNN_BiLSTM | 105,575 | 0.9117        | 0.0397 | 0.9714       | 0.1394        | 0.6286    | 22/35   |

Best on hold-out: **Baseline** · Best on CV: **BN\_GAP**



## Training curves (final retrain on full training pool)



# Variant: Baseline

Conv1D(32,k=4)→Pool→Conv1D(64,k=4)→Pool→Flatten→Dense(64)→Dropout(0.3)→Softmax

|                          |                 |
|--------------------------|-----------------|
| Trainable params         | 113,255         |
| CV mean ± std            | 0.9346 ± 0.0345 |
| Hold-out test acc / loss | 0.9714 · 0.0692 |
| Batch-test acc           | 0.7429 (26/35)  |

## Cross-validation folds

| Fold | Train (post-aug) | Val | Val acc | Best val | Final val |
|------|------------------|-----|---------|----------|-----------|
| 1    | 1464             | 62  | 0.9516  | 0.9516   | 0.9516    |
| 2    | 1470             | 61  | 0.9344  | 0.9836   | 0.9344    |
| 3    | 1470             | 61  | 0.9672  | 1.0000   | 0.9672    |
| 4    | 1470             | 61  | 0.9508  | 0.9508   | 0.9508    |
| 5    | 1470             | 61  | 0.8689  | 0.9180   | 0.8689    |

## Per-class hold-out metrics

| Class                | Precision | Recall | F1    | Support |
|----------------------|-----------|--------|-------|---------|
| Double_Lower         | 1.000     | 1.000  | 1.000 | 5       |
| Double_Nothing       | 1.000     | 0.800  | 0.889 | 5       |
| Double_Pistol_Recoil | 1.000     | 1.000  | 1.000 | 5       |
| Double_Raise         | 1.000     | 1.000  | 1.000 | 5       |
| Double_Wiggle        | 0.833     | 1.000  | 0.909 | 5       |
| Double_cmere         | 1.000     | 1.000  | 1.000 | 5       |
| Drum_Roll            | 1.000     | 1.000  | 1.000 | 5       |
| macro avg            | 0.976     | 0.971  | 0.971 | 35      |
| weighted avg         | 0.976     | 0.971  | 0.971 | 35      |

## Batch test per class (12\_Marcus)

| Class                | Correct | Total | Accuracy |
|----------------------|---------|-------|----------|
| Double_Lower         | 5       | 5     | 1.000    |
| Double_Nothing       | 1       | 5     | 0.200    |
| Double_Pistol_Recoil | 4       | 5     | 0.800    |
| Double_Raise         | 5       | 5     | 1.000    |
| Double_Wiggle        | 5       | 5     | 1.000    |
| Double_cmere         | 5       | 5     | 1.000    |
| Drum_Roll            | 1       | 5     | 0.200    |
| Overall              | 26      | 35    | 0.743    |

# Variant: Shallow

Single Conv block: Conv1D(32,k=5)→Pool→Flatten→Dense(32)→Dropout(0.3)→Softmax

|                          |                 |
|--------------------------|-----------------|
| Trainable params         | 72,487          |
| CV mean ± std            | 0.9216 ± 0.0379 |
| Hold-out test acc / loss | 0.9429 · 0.2130 |
| Batch-test acc           | 0.5714 (20/35)  |

## Cross-validation folds

| Fold | Train (post-aug) | Val | Val acc | Best val | Final val |
|------|------------------|-----|---------|----------|-----------|
| 1    | 1464             | 62  | 0.9194  | 0.9355   | 0.9194    |
| 2    | 1470             | 61  | 0.8689  | 0.8689   | 0.8689    |
| 3    | 1470             | 61  | 0.9836  | 0.9836   | 0.9836    |
| 4    | 1470             | 61  | 0.9344  | 0.9344   | 0.9344    |
| 5    | 1470             | 61  | 0.9016  | 0.9016   | 0.9016    |

## Per-class hold-out metrics

| Class                | Precision | Recall | F1    | Support |
|----------------------|-----------|--------|-------|---------|
| Double_Lower         | 1.000     | 0.800  | 0.889 | 5       |
| Double_Nothing       | 1.000     | 0.800  | 0.889 | 5       |
| Double_Pistol_Recoil | 1.000     | 1.000  | 1.000 | 5       |
| Double_Raise         | 1.000     | 1.000  | 1.000 | 5       |
| Double_Wiggle        | 0.714     | 1.000  | 0.833 | 5       |
| Double_cmere         | 1.000     | 1.000  | 1.000 | 5       |
| Drum_Roll            | 1.000     | 1.000  | 1.000 | 5       |
| macro avg            | 0.959     | 0.943  | 0.944 | 35      |
| weighted avg         | 0.959     | 0.943  | 0.944 | 35      |

## Batch test per class (12\_Marcus)

| Class                | Correct | Total | Accuracy |
|----------------------|---------|-------|----------|
| Double_Lower         | 5       | 5     | 1.000    |
| Double_Nothing       | 1       | 5     | 0.200    |
| Double_Pistol_Recoil | 0       | 5     | 0.000    |
| Double_Raise         | 5       | 5     | 1.000    |
| Double_Wiggle        | 3       | 5     | 0.600    |
| Double_cmere         | 5       | 5     | 1.000    |
| Drum_Roll            | 1       | 5     | 0.200    |
| Overall              | 20      | 35    | 0.571    |

# Variant: Deep

Three Conv blocks: Conv1D(32,4)→Conv1D(64,4)→Conv1D(128,3)→Flatten→Dense(128)→Dropout(0.4)→Softmax

|                          |                 |
|--------------------------|-----------------|
| Trainable params         | 204,007         |
| CV mean ± std            | 0.9380 ± 0.0239 |
| Hold-out test acc / loss | 0.9714 · 0.1302 |
| Batch-test acc           | 0.8571 (30/35)  |

## Cross-validation folds

| Fold | Train (post-aug) | Val | Val acc | Best val | Final val |
|------|------------------|-----|---------|----------|-----------|
| 1    | 1464             | 62  | 0.9194  | 0.9516   | 0.9194    |
| 2    | 1470             | 61  | 0.9672  | 0.9836   | 0.9672    |
| 3    | 1470             | 61  | 0.9672  | 0.9836   | 0.9672    |
| 4    | 1470             | 61  | 0.9180  | 0.9836   | 0.9180    |
| 5    | 1470             | 61  | 0.9180  | 0.9180   | 0.9180    |

## Per-class hold-out metrics

| Class                | Precision | Recall | F1    | Support |
|----------------------|-----------|--------|-------|---------|
| Double_Lower         | 1.000     | 1.000  | 1.000 | 5       |
| Double_Nothing       | 1.000     | 0.800  | 0.889 | 5       |
| Double_Pistol_Recoil | 1.000     | 1.000  | 1.000 | 5       |
| Double_Raise         | 1.000     | 1.000  | 1.000 | 5       |
| Double_Wiggle        | 0.833     | 1.000  | 0.909 | 5       |
| Double_cmere         | 1.000     | 1.000  | 1.000 | 5       |
| Drum_Roll            | 1.000     | 1.000  | 1.000 | 5       |
| macro avg            | 0.976     | 0.971  | 0.971 | 35      |
| weighted avg         | 0.976     | 0.971  | 0.971 | 35      |

## Batch test per class (12\_Marcus)

| Class                | Correct | Total | Accuracy |
|----------------------|---------|-------|----------|
| Double_Lower         | 5       | 5     | 1.000    |
| Double_Nothing       | 2       | 5     | 0.400    |
| Double_Pistol_Recoil | 5       | 5     | 1.000    |
| Double_Raise         | 5       | 5     | 1.000    |
| Double_Wiggle        | 5       | 5     | 1.000    |
| Double_cmere         | 5       | 5     | 1.000    |
| Drum_Roll            | 3       | 5     | 0.600    |
| Overall              | 30      | 35    | 0.857    |

# Variant: BN\_GAP

Conv+BN+ReLU ×3 with GlobalAveragePooling1D head — fewer params, less overfit

|                          |                 |
|--------------------------|-----------------|
| Trainable params         | 57,319          |
| CV mean ± std            | 0.9673 ± 0.0232 |
| Hold-out test acc / loss | 0.9714 · 0.1233 |
| Batch-test acc           | 0.7143 (25/35)  |

## Cross-validation folds

| Fold | Train (post-aug) | Val | Val acc | Best val | Final val |
|------|------------------|-----|---------|----------|-----------|
| 1    | 1464             | 62  | 0.9677  | 0.9677   | 0.9677    |
| 2    | 1470             | 61  | 0.9836  | 0.9836   | 0.9836    |
| 3    | 1470             | 61  | 1.0000  | 1.0000   | 1.0000    |
| 4    | 1470             | 61  | 0.9508  | 0.9672   | 0.9508    |
| 5    | 1470             | 61  | 0.9344  | 0.9508   | 0.9344    |

## Per-class hold-out metrics

| Class                | Precision | Recall | F1    | Support |
|----------------------|-----------|--------|-------|---------|
| Double_Lower         | 1.000     | 1.000  | 1.000 | 5       |
| Double_Nothing       | 1.000     | 0.800  | 0.889 | 5       |
| Double_Pistol_Recoil | 1.000     | 1.000  | 1.000 | 5       |
| Double_Raise         | 1.000     | 1.000  | 1.000 | 5       |
| Double_Wiggle        | 0.833     | 1.000  | 0.909 | 5       |
| Double_cmere         | 1.000     | 1.000  | 1.000 | 5       |
| Drum_Roll            | 1.000     | 1.000  | 1.000 | 5       |
| macro avg            | 0.976     | 0.971  | 0.971 | 35      |
| weighted avg         | 0.976     | 0.971  | 0.971 | 35      |

## Batch test per class (12\_Marcus)

| Class                | Correct | Total | Accuracy |
|----------------------|---------|-------|----------|
| Double_Lower         | 5       | 5     | 1.000    |
| Double_Nothing       | 1       | 5     | 0.200    |
| Double_Pistol_Recoil | 4       | 5     | 0.800    |
| Double_Raise         | 5       | 5     | 1.000    |
| Double_Wiggle        | 5       | 5     | 1.000    |
| Double_cmere         | 5       | 5     | 1.000    |
| Drum_Roll            | 0       | 5     | 0.000    |
| Overall              | 25      | 35    | 0.714    |

# Variant: WideKernel

Conv1D(32,k=8)→Pool→Conv1D(64,k=8)→Pool→Flatten→Dense(64)→Dropout(0.3)→Softmax

|                          |                 |
|--------------------------|-----------------|
| Trainable params         | 131,687         |
| CV mean ± std            | 0.9249 ± 0.0304 |
| Hold-out test acc / loss | 0.9429 · 0.1480 |
| Batch-test acc           | 0.6857 (24/35)  |

## Cross-validation folds

| Fold | Train (post-aug) | Val | Val acc | Best val | Final val |
|------|------------------|-----|---------|----------|-----------|
| 1    | 1464             | 62  | 0.9194  | 0.9516   | 0.9194    |
| 2    | 1470             | 61  | 0.9508  | 0.9672   | 0.9508    |
| 3    | 1470             | 61  | 0.9672  | 0.9672   | 0.9672    |
| 4    | 1470             | 61  | 0.9016  | 0.9344   | 0.9016    |
| 5    | 1470             | 61  | 0.8852  | 0.9180   | 0.8852    |

## Per-class hold-out metrics

| Class                | Precision | Recall | F1    | Support |
|----------------------|-----------|--------|-------|---------|
| Double_Lower         | 1.000     | 1.000  | 1.000 | 5       |
| Double_Nothing       | 0.800     | 0.800  | 0.800 | 5       |
| Double_Pistol_Recoil | 1.000     | 0.800  | 0.889 | 5       |
| Double_Raise         | 1.000     | 1.000  | 1.000 | 5       |
| Double_Wiggle        | 0.833     | 1.000  | 0.909 | 5       |
| Double_cmere         | 1.000     | 1.000  | 1.000 | 5       |
| Drum_Roll            | 1.000     | 1.000  | 1.000 | 5       |
| macro avg            | 0.948     | 0.943  | 0.943 | 35      |
| weighted avg         | 0.948     | 0.943  | 0.943 | 35      |

## Batch test per class (12\_Marcus)

| Class                | Correct | Total | Accuracy |
|----------------------|---------|-------|----------|
| Double_Lower         | 5       | 5     | 1.000    |
| Double_Nothing       | 0       | 5     | 0.000    |
| Double_Pistol_Recoil | 0       | 5     | 0.000    |
| Double_Raise         | 5       | 5     | 1.000    |
| Double_Wiggle        | 5       | 5     | 1.000    |
| Double_cmere         | 5       | 5     | 1.000    |
| Drum_Roll            | 4       | 5     | 0.800    |
| Overall              | 24      | 35    | 0.686    |

# Variant: CNN\_LSTM

Conv1D(32,4)→Pool→Conv1D(64,4)→Pool→Dropout→LSTM(64)→Dense(64)→Dropout→Softmax

|                          |                 |
|--------------------------|-----------------|
| Trainable params         | 68,455          |
| CV mean ± std            | 0.9117 ± 0.0494 |
| Hold-out test acc / loss | 0.8857 · 0.3586 |
| Batch-test acc           | 0.4857 (17/35)  |

## Cross-validation folds

| Fold | Train (post-aug) | Val | Val acc | Best val | Final val |
|------|------------------|-----|---------|----------|-----------|
| 1    | 1464             | 62  | 0.9355  | 0.9355   | 0.9355    |
| 2    | 1470             | 61  | 0.8197  | 0.8197   | 0.8197    |
| 3    | 1470             | 61  | 0.9672  | 0.9672   | 0.9672    |
| 4    | 1470             | 61  | 0.9180  | 0.9180   | 0.9180    |
| 5    | 1470             | 61  | 0.9180  | 0.9180   | 0.9180    |

## Per-class hold-out metrics

| Class                | Precision | Recall | F1    | Support |
|----------------------|-----------|--------|-------|---------|
| Double_Lower         | 0.714     | 1.000  | 0.833 | 5       |
| Double_Nothing       | 1.000     | 0.600  | 0.750 | 5       |
| Double_Pistol_Recoil | 1.000     | 1.000  | 1.000 | 5       |
| Double_Raise         | 1.000     | 1.000  | 1.000 | 5       |
| Double_Wiggle        | 1.000     | 0.600  | 0.750 | 5       |
| Double_cmere         | 0.833     | 1.000  | 0.909 | 5       |
| Drum_Roll            | 0.833     | 1.000  | 0.909 | 5       |
| macro avg            | 0.912     | 0.886  | 0.879 | 35      |
| weighted avg         | 0.912     | 0.886  | 0.879 | 35      |

## Batch test per class (12\_Marcus)

| Class                | Correct | Total | Accuracy |
|----------------------|---------|-------|----------|
| Double_Lower         | 5       | 5     | 1.000    |
| Double_Nothing       | 0       | 5     | 0.000    |
| Double_Pistol_Recoil | 0       | 5     | 0.000    |
| Double_Raise         | 5       | 5     | 1.000    |
| Double_Wiggle        | 0       | 5     | 0.000    |
| Double_cmere         | 5       | 5     | 1.000    |
| Drum_Roll            | 2       | 5     | 0.400    |
| Overall              | 17      | 35    | 0.486    |



# Variant: CNN\_BiLSTM

Conv1D(32,4)→Pool→Conv1D(64,4)→Pool→Dropout→BiLSTM(64)→Dense(64)→Dropout→Softmax

|                          |                 |
|--------------------------|-----------------|
| Trainable params         | 105,575         |
| CV mean ± std            | 0.9117 ± 0.0397 |
| Hold-out test acc / loss | 0.9714 · 0.1394 |
| Batch-test acc           | 0.6286 (22/35)  |

## Cross-validation folds

| Fold | Train (post-aug) | Val | Val acc | Best val | Final val |
|------|------------------|-----|---------|----------|-----------|
| 1    | 1464             | 62  | 0.9194  | 0.9194   | 0.9194    |
| 2    | 1470             | 61  | 0.9508  | 0.9672   | 0.9508    |
| 3    | 1470             | 61  | 0.9344  | 0.9836   | 0.9344    |
| 4    | 1470             | 61  | 0.8361  | 0.9180   | 0.8361    |
| 5    | 1470             | 61  | 0.9180  | 0.9180   | 0.9180    |

## Per-class hold-out metrics

| Class                | Precision | Recall | F1    | Support |
|----------------------|-----------|--------|-------|---------|
| Double_Lower         | 1.000     | 1.000  | 1.000 | 5       |
| Double_Nothing       | 1.000     | 0.800  | 0.889 | 5       |
| Double_Pistol_Recoil | 1.000     | 1.000  | 1.000 | 5       |
| Double_Raise         | 1.000     | 1.000  | 1.000 | 5       |
| Double_Wiggle        | 0.833     | 1.000  | 0.909 | 5       |
| Double_cmere         | 1.000     | 1.000  | 1.000 | 5       |
| Drum_Roll            | 1.000     | 1.000  | 1.000 | 5       |
| macro avg            | 0.976     | 0.971  | 0.971 | 35      |
| weighted avg         | 0.976     | 0.971  | 0.971 | 35      |

## Batch test per class (12\_Marcus)

| Class                | Correct | Total | Accuracy |
|----------------------|---------|-------|----------|
| Double_Lower         | 5       | 5     | 1.000    |
| Double_Nothing       | 1       | 5     | 0.200    |
| Double_Pistol_Recoil | 0       | 5     | 0.000    |
| Double_Raise         | 5       | 5     | 1.000    |
| Double_Wiggle        | 5       | 5     | 1.000    |
| Double_cmere         | 5       | 5     | 1.000    |
| Drum_Roll            | 1       | 5     | 0.200    |
| Overall              | 22      | 35    | 0.629    |